Compton scattering of 662-keV γ rays from a steel target: energy–angle relation, electron rest energy, and differential cross section

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Abstract

I. INTRODUCTION

Introduction

II. THEORY AND EXPECTED RESULTS

A. Kinematics of Compton scattering

B. Thomson and Klein–Nishina cross sections

C. Interaction of the scattered photons with the target and the detector

D. From counts to a cross section

E. Expected results

III. EXPERIMENTAL SETUP AND DATA

A. Apparatus

B. Data sets

IV. ANALYSIS

The analysis is implemented in Python. A single configuration file holds every input and its uncertainty, and one script regenerates all numbers, tables and figures of this report.

A. Energy calibration and resolution

B. The measured spectra

C. The primary beam

D. Target thickness

E. Detector efficiency

F. Forward-folding model of the scattered spectrum

G. Signal extraction

V. UNCERTAINTIES AND ERRORS

A. Statistical uncertainties

B. Signal-extraction uncertainty

C. Systematic uncertainties

D. Correlations and combination

E. Energy uncertainties

F. Errors identified and corrected

G. Compton kinematics and the electron rest energy

H. Differential cross section

I. Dependence on the target assumptions

VI. DISCUSSION

VII. CONCLUSION

Conclusion

Appendix A: Calibration cross-check

Appendix B: Reproducing the results